

Afficionado Coffee Roasters 2025

Data Ingestion & Validation

The transaction dataset was first loaded into the analysis environment for further processing. Before performing any analysis, the data was validated to ensure its accuracy, consistency, and reliability.

The following validation steps were performed:

- 1. Load Transaction Dataset:**
The coffee shop transaction dataset was imported for analysis.
- 2. Validate Timestamp Formats:**
Transaction date and time fields were checked to ensure that they were stored in the correct and consistent format.
- 3. Check Missing and Duplicate Transaction IDs:**
The dataset was examined for missing transaction IDs and duplicate records to avoid inaccurate calculations.
- 4. Verify Logical Consistency:**
Transaction quantities and unit prices were checked to ensure that they contained valid positive values and did not contain logically incorrect entries.

These validation steps helped ensure that the dataset was clean and reliable before proceeding with sales trend and time-based performance analysis.

Feature Engineering (Temporal)

To perform time-based sales analysis, several temporal features were derived from the transaction date and time fields. These features helped identify sales patterns, peak transaction periods, and customer activity across different times of the day.

The following features were created:

- 1. Revenue per Transaction:**
Revenue was calculated by multiplying the transaction quantity by the unit price.

Revenue = transaction_qty × unit_price
- 2. Hour of Day:**
The transaction time was used to extract the hour of the day, ranging from **0 to 23**.
- 3. Day of Week:**
The transaction date was used to identify the day of the week, from **Monday to Sunday**.
- 4. Time Buckets:**
Transactions were grouped into four time-based categories:
 - **Morning:** 6:00–11:59

- **Afternoon:** 12:00–16:59
- **Evening:** 17:00–21:59
- **Late Hours:** 22:00–5:59

These derived features enabled detailed analysis of hourly sales trends, transaction volume, and customer activity across different time periods.

Sales Trend Analysis

Sales trends were analyzed to understand how revenue and transaction activity changed over time. The analysis focused on identifying daily and weekly patterns and comparing sales performance across different store locations.

The following analyses were performed:

1. **Daily Revenue Trends:**
Daily revenue was calculated and analyzed to identify changes in sales performance over time.
2. **Weekly Revenue and Transaction Analysis:**
Revenue and the number of transactions were aggregated on a weekly basis to understand broader sales patterns.
3. **Identification of Upward and Downward Patterns:**
Sales trends were examined to identify periods of increasing or decreasing revenue and transaction activity.
4. **Store-Level Trend Comparison:**
Sales trends across different store locations were compared to identify differences in performance and temporal sales patterns.

This analysis helped identify sales fluctuations, recurring patterns, and differences in store-level performance, supporting better operational and staffing decisions.

Day-of-Week Performance Analysis

Day-of-week analysis was performed to understand how customer activity and sales performance varied across different days of the week. This analysis helped identify high-performing and low-performing days and distinguish between weekday and weekend behaviour.

The following analyses were performed:

1. **Average Revenue by Day of Week:**
The average daily revenue was calculated for each day from Monday to Sunday to identify the days with higher and lower sales performance.
2. **Average Transaction Count by Day:**
The average number of transactions was calculated for each day of the week to measure customer activity.

Time-of-Day Demand Analysis

Time-of-day analysis was performed to understand how customer demand and revenue varied throughout the day. Hourly transaction and revenue patterns were analyzed to identify periods of high and low customer activity.

The following analyses were performed:

1. **Hourly Transaction Volume:**
The number of transactions was aggregated for each hour of the day to identify variations in customer demand.
2. **Hourly Revenue Distribution:**
Revenue was analyzed across different hours to determine which time periods generated the highest and lowest sales.
3. **Identification of Morning Rush Hours:**
Peak transaction periods during the morning were identified to understand early-day customer demand.
4. **Identification of Midday Slow Periods:**
Hours with relatively lower transaction volumes and revenue during the middle of the day were identified.
5. **Identification of Evening Peaks:**
Evening hours with increased transaction activity and revenue were analyzed to identify potential demand peaks.

The results of this analysis can help in optimizing staff allocation, inventory availability, and operational planning according to hourly customer demand.

3. **Weekday vs Weekend Comparison:**
Revenue and transaction activity on weekdays were compared with weekends to identify differences in sales and customer behaviour.
4. **Behavioural Interpretation:**
The observed patterns were interpreted in the context of workday and leisure-day behaviour, helping to understand how customer routines may influence coffee shop visits and purchasing activity.

This analysis provided insights into weekly customer patterns and helped support decisions related to staffing, inventory planning, and operational scheduling.

Cross-Location Temporal Comparison

A cross-location temporal analysis was performed to compare customer activity and sales patterns across different store locations. The objective was to identify similarities and differences in demand patterns based on time of day.

The following analyses were performed:

1. **Hourly Heatmaps per Store:**

Hourly transaction activity was visualized for each store using heatmaps to identify periods of high and low customer demand.

2. **Peak-Hour Alignment or Divergence:**

Peak transaction hours were compared across locations to determine whether stores experienced demand peaks at similar times or followed different patterns.

3. **Location-Specific Customer Behavior Insights:**

Differences in customer activity across locations were analyzed to identify location-specific behavioral patterns and demand characteristics.

This comparison helped identify whether staffing and operational strategies should be standardized across stores or adjusted according to the demand patterns of individual locations.

